AgriTalk: Calibrated Conversational Orchestration for Human-Supervised Agricultural Robots

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Abstract. Current IoRT (Internet of Robotic Things) control interfaces presuppose programming-level expertise, excluding the agronomists who are legally accountable for field decisions. This proposal, **AgriTalk** proposes a calibrated conversational orchestration layer: four scientifically grounded contributions address (i) Natural Language (NL) intent compilation with statistically principled conformal prediction uncertainty and Human-in-the-Loop (HITL) trigger policies, (ii) verifiably faithful explanation through Belief Vector Field geometry – the candidate's own prior work – rather than unreliable Chain-of-Thought, (iii) atomic temporal grounding against live IoT field state via streaming aggregation, and (iv) an empirical collaborative trust evaluation framework. Together they produce a human-auditable, GDPR-aligned, Metaflow-reproducible conversational layer for precision pesticide spraying—with non-bypassable safety gates.

Introduction and Research Context

Programme context. Agricultural robots (weeding, spraying, seeding, harvesting, autonomous tractors, and drones) cooperate in multi-modal IoRT systems requiring IoT data collection, streaming analytics, computer vision, cloud computing, and AI. Yet current platforms presuppose programmer-level expertise: farmers and agronomists – who are legally accountable for pesticide use, spray drift compliance, and crop safety – cannot directly command or supervise the systems they depend on (Chen et al., 2021; Wagner et al., 2025). **AgriTalk** addresses this by creating a unified conversational layer where end-users express high-level agricultural tasks in natural language, dynamically mapped to calibrated, explainable, and auditable IoRT control processes – transforming robotic spraying from reactive and ad hoc to knowledge-driven, safe, and operator-supervised.

Four scientific gaps driving AgriTalk: (G1) NL disambiguation: agricultural commands are contextually under-specified and temporally conditional – a single utterance conflates spatial reference, dosage, chemical identity, phenological timing, and weather constraints, requiring calibrated disambiguation beyond slot-filling NLU (Chen et al., 2021). **(G2) Calibration and explanation:** LLM softmax confidence is systematically overconfident on out-of-distribution inputs (Guo et al., 2017); Chain-of-Thought rationales are behaviourally generated, not computationally causal – anthropomorphizing them as transparent cognition produces dangerously misleading safety evaluations (Turpin et al., 2023; Kambhampati et al., 2025). **(G3) Temporal IoT grounding:** heterogeneous field streams must be atomically consistent for command grounding – no system currently solves this at agronomic granularity (Vargas-Solar et al., 2021). **(G4) Trust calibration:** the causal link between faithful explanation and operator trust in HITL[1] Agricultural systems are empirically unmeasured.

Focal Scenario. *"Spray more pesticide on the struggling section near the east boundary."* Simultaneous ambiguities: spatial reference (which east boundary?), dosage baseline (more than what?), chemical formulation (resistance profiles + phenological stage), spray drift legality (current wind speed). Wrong interpretation: crop destruction, regulatory violation. Over-confirmation: alert fatigue. AgriTalk resolves this with calibrated statistical confidence, verifiable attribution, and an immutable audit trail.

[1]HITL: Human-in-the-Loop – deliberate inclusion of human judgment before robot actuation.

Research Questions and Contributions

Research question breakdown (objective, data, expected output, PhD contribution):

RQ	Objective	Input Data	Expected Output	PhD Contribution
RQ1	Derive statistically valid HITL trigger policies from conformal NL intent calibration under seasonal distribution shift	AgroNLP corpus (labelled commands); seasonal calibration holdouts; conformal scores	Coverage-certified $\mathcal{C}(x)$; adaptive HITL policy per action class; ECE vs. softmax baseline	C1: first principled uncertainty-to-HITL mapping for safety-critical agricultural NL commands
RQ2	Provide verifiably faithful attribution by tracing BVF[2] geometry, not generating CoT narratives	Fine-tuned LLM layer activations; IG, LRP, SHAP, rollout, BVF; operator study (N$\geq$30)	BVF layer traces; cross-method Kendall τ; trust calibration gap Δ(C vs. B)	C2: BVF as computationally causal alternative to CoT—empirically tested not assumed
RQ3	Enable real-time atomic NL grounding against IoT field state; characterize failure under sensor degradation	Kafka streams (sensor, NDVI, weather, telemetry); fault injection; Gazebo simulation	Time-indexed field-state register; recall vs. degradation curve; V2 staleness alerts	C3: first atomic conversational grounding layer for agricultural IoRT at stream granularity
RQ4	Measure whether faithful attribution causally improves trust calibration vs. CoT; build reusable evaluation protocol	3-condition user study; HITL trace logs; trust surveys; Metaflow seasonal replay	CTEF protocol; trust gap Δ; seasonal reproducibility archive NASA-TLX	C4: closes empirical loop between explanation faithfulness and operator trust calibration

The following are my four expected research contributions for each of the research question:

- **C1 [AI Safety] Calibrated Intent Compiler (CIC):** Conformal prediction-based NL-to-command compilation with statistically guaranteed HITL trigger policies (EMNLP/ACL 2027).
- **C2 [Interpretability] Belief-Grounded Explanation System (BGES):** BVF layer-wise geometry for verifiably faithful attribution, validated by a human operator study (ACL/EMNLP 2028; FAccT 2029).
- **C3 [Streaming NLP] Temporal Streaming Grounding Aggregator (TSGA):** Atomically consistent time-indexed field-state register for contextual grounding under sensor uncertainty (Computers & Electronics in Agriculture Q1 2028 or VLDB 2028).
- **C4 [Collaborative Trust] Collaborative Trust Evaluation Framework (CTEF):** Mixed-methods framework measuring how calibrated uncertainty and faithful attribution causally improve operator trust (CHI 2029 or FAccT 2029).

System Architecture: AgriTalk

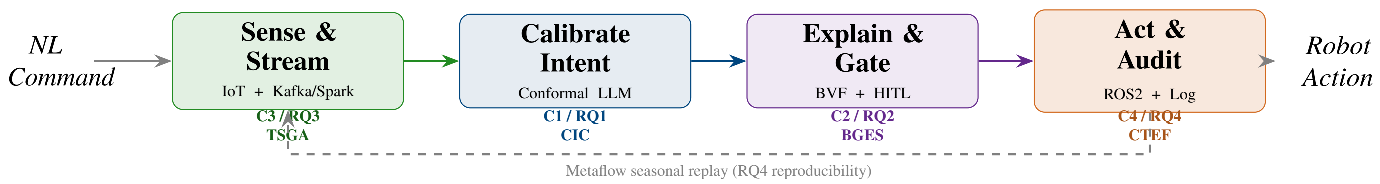


Figure 1: **AgriTalk: four-stage conversational orchestration pipeline (Figure 1).** Each stage maps to one research contribution and one information boundary group. The dashed arc is the Metaflow artifact replay path: archived sensor snapshots and model checkpoints from Year 1 (Lyon) can be re-executed on Year 2 (Milan) context, making RQ4 trust measurements reproducible and season-comparable without re-running real experiments.

Information boundaries and verifier layer. Acronyms: NDVI = Normalized Difference Vegetation Index (crop health; value $<$0.35 indicates stress zone); DQ = Data Quality (V1 verifier: freshness, completeness, cross-modality consistency); JWT = JSON Web Token (cryptographically signed auth); RBAC = Role-Based Access Control; ROS2 = Robot Operating System 2 (DDS middleware); KB = Knowledge Base (EPPO/PPDB agronomic records: spray drift limits, dosage protocols, resistance profiles). AgriTalk crosses five information

[2]BVF: Belief Vector Field—layer-wise geometric trajectory of model representational state (Saha, 2026).

boundaries (B1–B5), each guarded by a dedicated verifier (V1–V5). The HITL gate (V5) is architecturally non-bypassable: five trigger conditions—action type in {SPRAY, ABORT, DOSAGE_CHANGE, ZONE_OVERRIDE, EMERGENCY_STOP}, $|\mathcal{C}(x)| \geq 2$, unverified KB entity, DQ failure, or role = OBSERVER—must be cleared before robot actuation. No API pathway overrides this.

INPUTS NL command from agronomist operator | IoT sensors: NDVI, wind, soil-moisture, GPS, spray telemetry | KB: EPPO/PPDB agronomic constraints (dosage, resistance, drift limits)

B1+B2 | Sense & Stream [C3 / TSGA] Kafka topics: `sensor.raw` (15 s) · `ndvi.updates` (per drone mission) · `weather` (10 min) · `robot.telemetry` (1 s) · `spray.events` (immutable audit) $\to$ Spark 15-min rolling aggregates $\to$ time-indexed field-state register *V1 DQ Verifier | V2 Staleness Verifier*

B3 | Language $\to$ Intent [C1 / CIC] LoRA fine-tuned MBZUAI agri-LLM (8 intent classes: SPRAY, ABORT, DOSAGE_CHANGE, QUERY, MONITOR, SCHEDULE, ZONE_OVERRIDE, EMERGENCY_STOP) $\to$ RAPS conformal wrapper: $\mathbb{P}(y \in \mathcal{C}(x)) \geq 0.95$ $|\mathcal{C}|{=}1$: proceed $|\mathcal{C}|{\geq}2$: HITL $|\mathcal{C}|{=}5$: reject *V3 Coverage Verifier (α=0.05)*

B4 | Intent $\to$ Command [C2 / BGES] BVF layer curvature + torsion $\to$ IG + LRP + SHAP + Attention Rollout cross-validated by Kendall $\tau{>}0.5$ $\to$ V4 sufficiency test (Δacc$>$ 0.2 on top-k removal) $\to$ KB-RAG grounding (EPPO entity resolution) $\to$ rule-auditable explanation string + KB source URIs *V4 Attribution Verifier*

B5 | Command $\to$ Robot [C1+C2 / Safety Gate] V5 HITL gate (5 non-bypassable conditions) $\to$ JWT-authenticated command $\to$ RBAC role check (OBSERVER / OPERATOR / AGRONOMIST) $\to$ ROS2 DDS publish $\to$ immutable audit record: {run_id, $\mathcal{C}(x)$, V1–V5 states, attribution top-k, role, timestamp} *V5 HITL Safety Gate*

OUTPUTS Robot action (ROS2) | Faithful explanation (BVF trace + KB URI) | Immutable audit log (regulatory + Metaflow seasonal replay for C4) | HITL decision record

Figure 2: **AgriTalk 5-boundary system architecture (Figure 2).** Each horizontal tier is one information boundary. Color: green = streaming data management (C3); blue = calibrated intent (C1); purple = attribution/explanation (C2); orange = safety gate (C1+C2). Every tier emits a verifiable artifact; V5's immutable audit record is both the regulatory accountability anchor and the scientific data substrate for C4 (CTEF) seasonal trust measurement.

Methods and Experiments

C1 – Calibrated Intent Compiler (RQ1). The AgroNLP corpus is built via structured agronomist interviews (5–8 experts at UCBLyon1 partner farms) producing intent-labelled commands across 8 classes; the safety-critical `ABORT` class ($\approx$2% of natural distribution) is handled by stratified sampling and asymmetric focal loss, with seasonal cohort splitting (Lyon Year 1 $\to$ Milan Year 2) preventing temporal data leakage. LoRA fine-tuning adapts the MBZUAI agricultural LLM[3]; RAPS conformal wrapping (Angelopoulos and Bates, 2023) delivers per-class coverage $\mathbb{P}(y \in \mathcal{C}(x)) \geq 0.95$. *Core open question (RQ1):* does exchangeability hold under seasonal drift? Experiments: (a) coverage stability (Year 2 test vs. Year 1 calibration sets); (b) HITL policy ablation (conformal-triggered vs. softmax-threshold vs. always-HITL); (c) Gazebo fault injection. ECE[4] monitors calibration quality; V3 triggers recalibration when coverage degrades $>$0.02. Novel parameter extraction from EPPO/PPDB manuals (spray dosage, resistance profiles) is embedded as KB grounding—transforming robotic spraying from ad-hoc to knowledge-driven (GreenFieldData Consortium, 2026).

C2 – Belief-Grounded Explanation System (RQ2). CoT traces are behavioral outputs, not computational records—systematically unfaithful to the model's actual processing (Turpin et al., 2023; Kambhampati et al., 2025; Oxford Artificial Intelligence and Global Impact Institute, 2025). BVF (Saha, 2026) traces layer-wise representational geometry: syntactic (verb/object, early layers) $\to$ semantic ("struggling section" = NDVI$<$0.35?, mid layers) $\to$ agronomic safety (dosage within EPPO limits?, late layers)—a verifiable computational pathway. Five attribution methods (IG (Sundararajan et al., 2017), LRP (Bach et al., 2015), SHAP (Lundberg and Lee, 2017), Attention Rollout (Abnar and Zuidema, 2020), BVF) are cross-validated by Kendall τ and V4 sufficiency/necessity tests. A 3-condition between-subjects operator study (N$\geq$30, Year 2, UniMI): **A** no explanation; **B** CoT narrative; **C** faithful BVF + KB URI string. Primary measure: trust calibration gap Δ (perceived minus empirical reliability); analysis: Wilcoxon signed-rank, Cohen's d, NASA-TLX cognitive workload (faithful explanation must not overload operators). This realises the advertisement objective of stakeholder empowerment through natural

[3] `MBZUAI/agriculture-llm`, HuggingFace.
[4] ECE: average deviation between model confidence and empirical accuracy across bins (Guo et al., 2017).

language corrections aligned to expert knowledge, with human-AI co-evolution of robot behaviour (GreenFieldData Consortium, 2026).

C3 – Temporal Streaming Grounding Aggregator (RQ3). TSGA maintains a time-indexed field-state register: Kafka topics (`sensor.raw` 15 s; `ndvi.updates` per mission; `weather` 10 min; `telemetry` 1 s; `spray.events` immutable) with Avro schema registry; Spark Structured Streaming applies 15-min rolling aggregates. Grounding uses the exact register snapshot at query receipt time—resolving temporal ambiguity: "spray now" uses the live register; "was Tuesday safe?" uses the Metaflow-archived artifact from that timestamp. Failure boundary experiments: sensor dropout (10–50%), telemetry lag (>5 min), GPS/vision conflict—characterising connectivity-limited rural farm risk rather than assuming reliable IoT. This directly implements the advertisement's objective of conversational streaming dataflow orchestration (GreenFieldData Consortium, 2026).

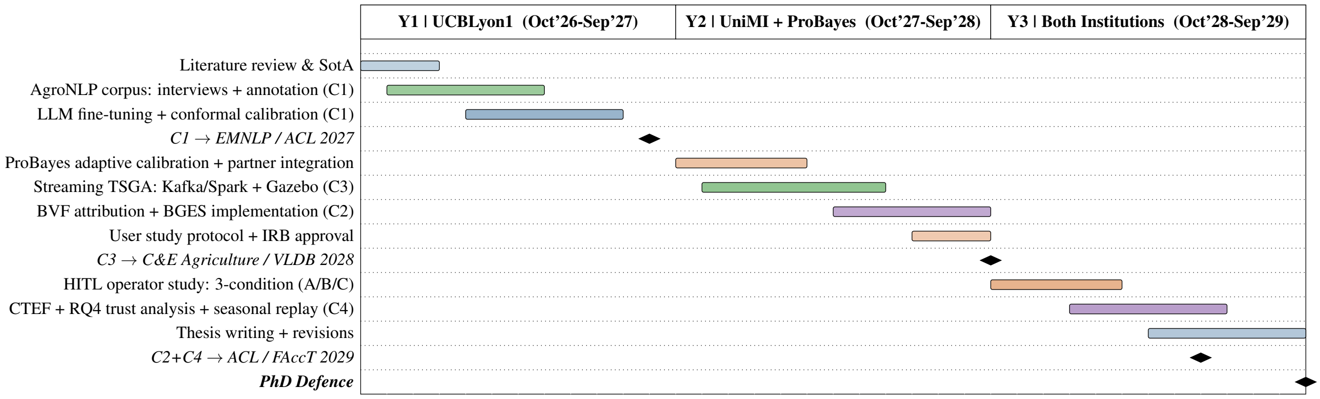
C4 – Collaborative Trust Evaluation Framework (RQ4). CTEF operationalises *collaborative trust*: the operator's calibrated confidence in AI outputs measured as willingness to act while retaining situational authority. Four instruments: (1) pre/post trust calibration survey; (2) behavioral trace analysis (HITL decisions per explanation condition); (3) think-aloud protocol; (4) Metaflow seasonal replay—re-running archived pipeline steps on historical snapshots to test cross-season policy generalization. Statistical analysis: Wilcoxon signed-rank, Cohen's d, regression controlling for role and experience. All pipeline artifacts are immutably versioned: {run_id, git_sha, artifact_hash, season_label, farm_id}—the scientific infrastructure making RQ4 answerable across seasons and farms. Three deployment tiers: *local* (UCBLyon1 GPU, fine-tuning), *edge* (NVIDIA Jetson AGX Orin, 8-bit quantized, P95 latency <800 ms), *cloud* (Azure/`@kubernetes`, federated C4 replay).

Evaluation Framework and Datasets

Metric	Justification	Target
Macro-F1 (intent)	Multi-class imbalance; ABORT rare but asymmetrically costly (Chen et al., 2021)	$\geq$ baseline
ECE (calibration)	Overconfidence vs. softmax (Guo et al., 2017); trust requires reliable confidence	$<$ softmax ECE
ABORT recall	Type-II error catastrophic; safety-critical class	Maximize
Coverage stability	Conformal drift under seasonal shift; recalibration trigger	Monitor
Kendall τ(IG,LRP)	Cross-method attribution agreement; faithfulness signal	$\tau > 0.5$
Trust calibration gap	Perceived $-$ empirical reliability; condition C vs. B	Δ(C$-$B)< 0
NASA-TLX workload	Faithful explanation must not overload practitioners	Non-inferior to B
Edge P95 latency	ABORT safety on Jetson AGX Orin deployment	<800 ms

Year	Institution	Datasets and Sources	Purpose
Y1	UCBLyon1	AgroNLP corpus (novel, agronomist interviews); PANGAEA soil/crop records; MBZUAI agri-LLM (HuggingFace, fine-tuning base); ACRE precision benchmark; OpenWeather Lyon 2020–2026 API	C1 corpus construction; C3 grounding baseline; LLM fine-tuning
Y2	UniMI + ProBayes	ProBayes partner farm logs (GDPR-compliant, federated); UniMI experimental spray records; FieldRobot Event 2026/2027 simulation; EPPO/PPDB live API; USDA-ARS-AgAID farmland_agriculture (HuggingFace)	C2 operator study; C3 failure boundary; real-robot validation
Y3	Both	Cross-farm federated corpus (2–4 farms, CFL (Saha, 2024)); synthetic adversarial corpus; Metaflow seasonal replay artifacts	C4 trust evaluation; generalization; CTEF reproducibility

Three-Year Roadmap and Expected Outcomes



Expected outcomes: (1) 3 peer-reviewed publications (C1/EMNLP 2027, C3/CEA-VLDB 2028, C2+C4/CHI-FAccT 2029); (2) open-source AgriTalk prototype (Metaflow pipelines, conformal module, BVF layer); (3) AgroNLP benchmark corpus (public release); (4) CTEF evaluation protocol (reusable beyond this project); (5) complete Metaflow artifact archive for seasonal replay.

Risk	**Primary Plan**	**Backup**
Real robot access delayed	Gazebo simulation (Months 1–18); all C1/C3 papers feasible on sim + ACRE	Robot needed only for C2 user study + real spray logs
Corpus too small	Agronomist interviews + expert-validated synthetic commands	MultiWOZ agricultural domain transfer
Conformal coverage drifts	Adaptive recalibration; per-season calibration checkpoints	Conservative fixed threshold; full HITL fallback
BVF over-smoothing at depth	Multi-layer torsion analysis; curvature monitoring	SHAP + LRP primary; BVF as supplementary analysis

Responsible AI, Design Choices, and Candidature

Responsible AI by design. (1) Safety-by-architecture: V5 HITL gate is non-bypassable; `ABORT` requires AGRONOMIST role; no code pathway can override it. **(2) Explainability before actuation:** V4 attribution verifier runs *before* command execution—explanation is a gate, not an audit footnote. **(3) Privacy / GDPR:** federated CFL (Saha, 2024) ensures raw farm data never leaves the farm; differential privacy on cross-farm aggregation. **(4) Failure transparency:** all V1–V5 failures surface with explicit, audit-logged reason codes (DQ channel, staleness duration, coverage shortfall, attribution disagreement, latency violation)—no silent degradation. **(5) Operator autonomy:** HITL design intentionally preserves human situational authority; AgriTalk assists—it does not automate away agronomist judgment.

Design rationale. *Conformal prediction over Bayesian calibration:* distribution-free coverage from a single calibration pass—tractable at 800 ms edge inference without MCMC (Angelopoulos and Bates, 2023; **?**). *BVF over CoT:* faithfulness is a testable empirical claim (RQ2), not a design preference. *Open models:* gradient access for attribution; reproducible on UCBLyon1 GPU without API costs. *Simulation-first:* Year 1 Gazebo → Year 2 real robot.

Candidature. My ongoing BVF research (Saha, 2026) provides direct grounding in layer-wise attribution and over-smoothing artifacts (C2); my CFL work (Saha, 2024) is the direct basis for Year 3 cross-farm generalization (C3 extension). Teaching AI and IoT to 50+ student projects built practical understanding of sensor deployment and non-specialist explanation design, essential for the C4 user study. Supervision aligns precisely: Dr. Vargas-Solar (streaming data management, DAG orchestration, conversational analytics) → C3/C4; Prof. Oberti (precision agriculture, field validation, spray robotics) → C1/C2 operator study. The open questions I find most compelling

– whether conformal coverage survives seasonal distribution drift and whether faithful attribution measurably improves trust calibration or merely overwhelms operators – are exactly why this is a PhD and not an engineering project.

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